

- 4 The airways of the gas exchange system are lined with epithelium. Gradual changes in the structural features of this epithelium occur as the airways branch and become increasingly narrow.
- (a) Table 4.1 shows the changes that occur in the number of goblet cells in the epithelium of the different structures of the gas exchange system.

Table 4.1

gas exchange structure	number of goblet cells in epithelium
trachea	many
bronchi	many
larger bronchioles	very few
smaller bronchioles	none
alveoli	none

Goblet cells produce mucus, which is important in maintaining the health of the airways.

The smallest bronchioles closest to the alveoli are known as respiratory bronchioles.

Suggest **and** explain why respiratory bronchioles do not have any goblet cells.

.....

.....

.....

.....

..... [2]

- (b) Fig. 4.1 is a photomicrograph of a section through a bronchiole, which is surrounded by alveoli.

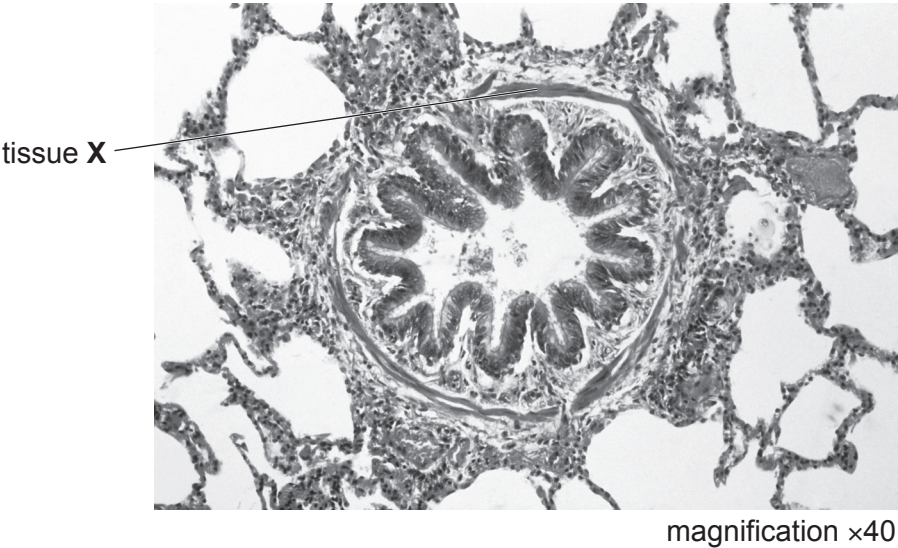


Fig. 4.1

- (i) There are structural differences between the epithelium of the bronchiole and the epithelium of an alveolus.

Describe the differences between the epithelium of bronchioles and the epithelium of alveoli, **other than** differences in the number of goblet cells.

.....

.....

.....

.....

.....

.....

.....

..... [3]

- (ii) Tissue **X**, shown in Fig. 4.1, is located in the wall of the bronchiole.

Name tissue **X** and outline the function of tissue **X** in the bronchiole.

tissue **X** =

.....

.....

.....

.....

.....

..... [3]

[Total: 8]